Wooseong Jeong

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Education

- **Korea Advanced Institute of Science and Technology (KAIST)** — Daejeon, Korea
  Ph.D. Candidate in Mechanical Engineering — March 2023 – February 2028 (Expected)
  Advisor: Kuk-Jin Yoon
  GPA: 3.81/4.3
- **Korea Advanced Institute of Science and Technology (KAIST)** — Daejeon, Korea
  M.S. in Mechanical Engineering — September 2021 – February 2023
  Advisor: Kuk-Jin Yoon
  GPA: 4.10/4.3
- **Korea Advanced Institute of Science and Technology (KAIST)** — Daejeon, Korea
  B.S. in Mechanical Engineering and Electrical Engineering (Double Major) — March 2015 – February 2021
  GPA: 3.76/4.3, cum laude

Research Interests

- **Foundation Model Optimization and Adaptation**
  - Multi-task, multi-objective, and non-convex optimization
  - Parameter-efficient adaptation, model merging, and knowledge distillation (including on-policy distillation)
  - Test-time adaptation and in-context learning
- **Embodied Intelligence**
  - Vision-language-action models and uncertainty-guided reasoning
  - Robust perception and motion planning for autonomous systems

Publications

- **Wooseong Jeong** and Kuk-Jin Yoon, "The Coupling Is the Lever: Cross-Group Geometry for Joint Optimization", **ICLR 2027** (Under Review).
- Giwon Lee*, **Wooseong Jeong***, and Kuk-Jin Yoon, "Context Distillation Distills More Than Skills: Navigating Skill Gains and Cross-Domain Retention with State-Shared Drift", **ICLR 2027** (Under Review).
- Giwon Lee*, **Wooseong Jeong***, and Kuk-Jin Yoon, "Beyond Context Selection: Consensus over Responses to Alternative Contexts", **ICLR 2027** (Under Review).
- **Wooseong Jeong***, Yujeong Chae*, and Kuk-Jin Yoon, "LiRA-Merge: Online Model Merging for Test-Time Modality Rebalancing in LiDAR–RADAR 3D Detection", **AAAI 2027** (Under Review).
- **Wooseong Jeong***, Giwon Lee*, Taewon Seo*, Seonae Jeon*, Jaeyoung Lee, Daehee Park, and Kuk-Jin Yoon, "CARMA: Cross-Domain Module Merging for Robust Motion Planning and End-to-End Driving", **T-IV** (Under Review).
- Hyemin Yang*, **Wooseong Jeong***, Giwon Lee*, and Kuk-Jin Yoon, "Triggering Generalist Reasoning via Predictive Uncertainty for Dual-System VLA", **NeurIPS 2026.**
- **Wooseong Jeong***, Wonyoung Lee*, and Kuk-Jin Yoon, "Preference-Aligned LoRA Merging: Preserving Subspace Coverage and Addressing Directional Anisotropy", **CVPR 2026.**
- Wonyoung Lee*, **Wooseong Jeong***, and Kuk-Jin Yoon, "Label-Free Cross-Task LoRA Merging with Null-Space Compression", **CVPR 2026.**
- **Wooseong Jeong***, Jegyeong Cho*, Youngho Yoon*, Jaeyoung Lee, and Kuk-Jin Yoon, "Stabilizing Multi-Task Latent Spaces: Recursive Refinement with Coordinators in Partially Labeled Learning", **IEEE Access 2026.**
- Giwon Lee*, **Wooseong Jeong***, Daehee Park, Jaewoo Jeong, and Kuk-Jin Yoon, "Interaction-Merged Motion Planning: Effectively Leveraging Diverse Motion Datasets for Robust Planning", **ICCV 2025 (Highlight).**
- **Wooseong Jeong** and Kuk-Jin Yoon, "Resolving Token-Space Gradient Conflicts: Token Space Manipulation for Transformer-Based Multi-Task Learning", **ICCV 2025.**
- **Wooseong Jeong***, Jegyeong Cho*, Youngho Yoon*, and Kuk-Jin Yoon, "Synchronizing Task Behavior: Aligning Multiple Tasks during Test-Time Training", **ICCV 2025.**
- Jihun Kim*, Hoyong Kwon*, Hyeokjun Kweon*, **Wooseong Jeong**, and Kuk-Jin Yoon, "DC-TTA: Divide-and-Conquer Framework for Test-Time Adaptation of Interactive Segmentation", **ICCV 2025.**

- **Wooseong Jeong***, Jihun Kim*, Hyeokjun Kweon*, and Kuk-Jin Yoon, "Multi-View 3D Scene Abstraction From Drone-Captured RGB Images", **IEEE Access 2025.**
- **Wooseong Jeong** and Kuk-Jin Yoon, "Selective Task Group Updates for Multi-Task Optimization", **ICLR 2025.**
- **Wooseong Jeong** and Kuk-Jin Yoon, "Quantifying Task Priority for Multi-Task Optimization", **CVPR 2024.**
- Hyeokjun Kweon*, Hyeonseong Kim*, Yoonsu Kang*, Youngho Yoon*, **Wooseong Jeong**, and Kuk-Jin Yoon, "Pixel-wise Warping for Deep Image Stitching", **AAAI 2023.**

Patents

- Jae Young Lee, Seung Uk Ahn, Young Woo Seo, Kuk-Jin Yoon, **Wooseong Jeong**, Je Gyeong Cho, and Young Ho Yoon, "Apparatus and Method for Learning Multiple Tasks in a Partial Labeling Environment", Republic of Korea Patent Application No. 10-2026-0089936, Filed May 18, 2026.
- Kuk-Jin Yoon, Hyeokjun Kweon, Yoonsu Kang, **Wooseong Jeong**, Hyeonseong Kim, and Hwi-Sung Park, "Apparatus and Method for Generating Image Dataset for Learning and Evaluation of Image Stitching", Republic of Korea Patent No. 10-2645640, Registered Mar. 5, 2024.
- Kuk-Jin Yoon, Hyeokjun Kweon, Yoonsu Kang, **Wooseong Jeong**, Hyeonseong Kim, and Hwi-Sung Park, "Apparatus and Method for Image Stitching", Republic of Korea Patent No. 10-2571530, Registered Aug. 23, 2023.

Research Experience

- **Project Leader** March 2025 – Present
  Core Technology Development for Autonomous Driving in Unstructured Off-Road Environments
  Funded by Hanwha Aerospace. Real-world deployment target in unstructured off-road terrain.
- **Project Leader** March 2024 – March 2025
  Unmanned Swarm CPS Research Lab
  Funded by Agency for Defense Development (ADD). Multi-agent perception and coordination for autonomous robotic systems.
- **Project Leader** March 2022 – February 2024
  Development of a 2kg-Class Small Electro-Optical Tracking System for Drones
  Funded by Institute of Civil-Military Technology Cooperation. Multi-modal (EO/IR) detection and tracking deployed on drone hardware.
- **Project Member** March 2021 – February 2022
  Unmanned Swarm CPS Research Lab
  Funded by Agency for Defense Development (ADD). Multi-agent perception research for autonomous robotic systems.

Honors and Awards

- **Qualcomm Innovation Fellowship Korea 2025 – Finalist** 2025
  Selected paper: "Interaction-Merged Motion Planning: Effectively Leveraging Diverse Motion Datasets for Robust Planning".

Academic Service

- **Reviewer**: NeurIPS (2024, 2025), ICLR (2025, 2026), CVPR (2025, 2026), ICML (2025), ICCV (2025), ECCV (2026), BMVC (2026), AAAI (2027), IEEE TMM (2026)

Skills

Languages	Python (proficient), C++ (proficient), MATLAB
ML Frameworks	PyTorch, JAX, NumPy, OpenCV
Efficient Training	PyTorch DDP, DeepSpeed, flash-attention, xformers
Hugging Face	Transformers, Diffusers, PEFT, Accelerate
Experiment & DevOps	Weights & Biases, TensorBoard, Docker, Git
Robotics	ROS, ROS2
Spoken Languages	Korean (native), English (fluent)